

AI-Assisted Subcontractor Management & Performance Monitoring

Business Problem

Construction companies work with many subcontractors. AI continuously evaluates their performance, predicts risks, and recommends the best vendors instead of relying on manual evaluation.

Scenario

The system integrates with ERP, Project Management, Primavera, Quality, Safety, and Finance systems.

Input

- Subcontractor master data
- Project schedules
- Contract details
- Daily progress reports
- Inspection reports
- Quality reports
- Safety incidents
- Invoices & payments
- Delay logs
- Engineer evaluations
- Client feedback

What the AI System Does

- Collect data from company systems.
- Clean and validate data.
- Calculate KPIs (Quality, Cost, Schedule, Safety).
- Generate subcontractor performance scores.
- Predict delays and contract breaches.
- Rank subcontractors.
- Recommend the best vendors for future projects.

Outputs

- Executive Dashboard
- Performance Scorecards
- Vendor Rankings
- Delay Risk Alerts
- Contract Breach Alerts
- Monthly KPI Reports
- Recommended Vendors

How Companies Use It

- Monitor subcontractors in real time.
- Select the best vendors.
- Reduce delays.
- Improve procurement decisions.
- Increase accountability.

Suggested AI Models

- Random Forest Regressor
- Random Forest/XGBoost Classifier
- Weighted Scoring
- Ranking Algorithm

Use Case Document

AI-Assisted Subcontractor Management & Performance Monitoring

1. Purpose

Provide an AI-driven solution that evaluates subcontractors, predicts risks, monitors KPIs, and recommends the best vendors for construction projects.

2. Actors

Procurement Manager

Project Manager

Site Engineer

Contracts Manager

Executive Management

ERP/Primavera System

AI Engine

3. Preconditions

- Company has registered subcontractors.
- Projects and contracts exist.
- ERP/Primavera and quality reports are integrated.

4. Input Data

- Vendor profile
- Contract details
- Project schedules
- Daily progress
- Inspection reports
- Quality reports
- Safety incidents
- Invoices
- Payments
- Engineer evaluations
- Client feedback

5. Main Flow

1. User opens Subcontractor Performance Dashboard.
2. System retrieves project and subcontractor data.
3. AI cleans and validates the data.
4. KPIs are calculated (Quality, Cost, Schedule, Safety).
5. ML model predicts delay probability.
6. ML model predicts contract breach risk.
7. AI calculates overall performance score.
8. AI ranks subcontractors.
9. Dashboard and reports are generated.
10. Managers receive alerts and recommendations.

6. AI Processing

- Data preprocessing
- Feature engineering
- Performance scoring model
- Delay prediction model
- Risk classification model
- Recommendation engine
- Dashboard generation

7. Outputs

- Performance scorecard
- Vendor ranking
- Delay risk alerts
- Contract breach alerts
- Recommended vendors
- Executive dashboard
- Monthly KPI reports

8. Exception Flows

Missing project data -> notify user.

Duplicate records -> merge automatically.

Invalid contract values -> send for review.

9. Business Rules

Quality weight: 35%

Schedule: 25%

Cost: 20%

Safety: 10%

Client Rating: 10%

10. Benefits

- Better vendor selection
- Early delay detection
- Reduced project risks

- Objective subcontractor evaluation
- Improved procurement decisions
- Higher project quality
- Lower project cost